JOSEPH JIN

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Experience

Software Engineering, Databricks **July 2024 - Present**

Foundation Model Serving Team January 2025 - Present

Tools - Scala, MySQL, Kubernetes, Spark, PromQL

- Implemented and released state synchronization service for new multitenant AI inference product, ensuring cross-region service consistency and enabling rollout of cross-region GPU orchestration capabilities required for public launch
- Led multi-month design and implementation of SOX-compliant, high-availability billing system for new batch and multitenant AI inference products, coordinating work across engineering and finance teams; implemented slot-based billing model that increased margins to 30% and established new standard for multitenant pricing calculations
- Built outage-proof ledger system tracking every capacity change against billed usage, developing custom Spark sliding-window algorithms to process high-volume data; designed for zero billing impact during service outages
- Created internal hackathon project to reuse idle CPU cores on GPU nodes for LLM workloads, reducing overhead from ~100+ unused CPU cores per 8xH100 node and demonstrating $4-10M+/yr. in potential savings based on CPU hosting COGS

Machine Learning Platform Team July 2024 - December 2024

Tools - Scala, MySQL, Kubernetes, Python

- Reduced costs and simplified cluster management by migrating customer deployments from Knative to custom auto-scaling resources, created custom deployment health monitoring tools, and improved migration service to unblock future migrations from common issues
- Implemented and rolled out tools for ML inference Docker images to enforce and monitor network policies, enabling 50+ enterprise customers with compliance requirements on egress control to use inference

Engineering Responsibilities

- Led critical incident response as on-call engineer, resolved high-priority service outages, interacted with customers for troubleshooting and customizing compute capacity, maintained 99.5% service uptime

Software Engineering Intern, Databricks **May 2023 - August 2023**

- Designed and implemented Kubernetes controllers for accelerating infrastructure rollouts on ML-inference compute clusters hosting 2,000+ customer deployments, reducing deployment time by 10 hours per rollout, using Go and Kubernetes controller-runtime SDK
- Developed blue green deployment scheme for small ML-serving types, ensuring zero downtime for 500+ additional enterprise customers during infrastructure changes, also using custom Kubernetes controllers

Researcher, Berkeley Artificial Intelligence Research Lab **February 2022 - December 2022**

- Worked with Professor Trevor Darrell and PhD candidate Zhuang Liu to show how accuracy and convergence of vision models can be improved by scheduling regularization strength during training (accepted to ICML 2023 arxiv.org/pdf/2303.01500.pdf)

Course Staff, UC Berkeley EECS Department **August 2022 - December 2022**

- Taught subjects such as vector calculus, PCA, SVD, convex optimization, and duality in office hours for EECS 127 (Optimization Models) ; answered questions on class forums; graded assignments; debugged projects and tests; proctored exams

Education

University of California, Berkeley **Class of 2024**

B.S. Electrical Engineering and Computer Sciences GPA: 3.9/4.0

Relevant Coursework - Machine Learning, Reinforcement Learning, Artificial Intelligence, Deep Learning, Computer Vision, Advanced Algorithms, Data Structures, Optimization Models, Operating Systems, Databases, Quantum Computing, Computer Architecture, Computer Security, Cryptography, Discrete Math, Probability, Structure of Computer Programs, Circuit Design and Theory I/II, Physics for Engineers

Skills

Languages: C, Python, Java, Scala, PromQL, Golang, Swift, Javascript, R, Matlab

Languages/Tools: Kubernetes, Spark, PyTorch, Scikit-learn, NumPy, Pandas, Docker, CVXPY, Git, AWS, MySQL, MongoDB

- **Eta Kappa Nu Honors Society** - top 20% of Berkeley EECS undergraduate students by GPA
- **Gold Division, USA Computing Olympiad -** top 6% of competitors in the United States